

1. Soru

$$y(n) = \sum_{k=-\infty}^{\infty} x(k) \cdot h(n-k) \implies y(n) = \sum_{k=-\infty}^{\infty} \left(\frac{1}{3}\right)^k \cdot u(k-1) \cdot u(n-k+2)$$

$$y(n) = \sum_{k=1}^{\infty} \left(\frac{1}{3}\right)^k \cdot u(n-k+2) \quad n-k+2 \geq 0 \quad n+2 \geq k$$

$$y(n) = \sum_{k=1}^{n+2} \left(\frac{1}{3}\right)^k = \sum_{k=0}^{n+2} \left(\frac{1}{3}\right)^k - 1 = \frac{1 - \left(\frac{1}{3}\right)^{n+3}}{1 - \frac{1}{3}} - 1$$

$$y(n) = \frac{3 - \left(\frac{1}{3}\right)^{n+3} \cdot 3}{2} - 1 = \left[\frac{1 - \left(\frac{1}{3}\right)^{n+2}}{2} \right] \cdot u(n)$$

2. Soru

$$y(n) - 5y(n-1) + 6y(n-2) = x(n) \quad x(n) = \cos \frac{n\pi}{2} \quad y(-1) = y(-2) = 1$$

a) Doğal Çözüm ($y_d(n)$)

$$y(n) - 5y(n-1) + 6y(n-2) = 0$$

$$y(n) = \lambda^n \implies \lambda^n - 5\lambda^{n-1} + 6\lambda^{n-2} = 0$$

$$\lambda^{n-2}(\lambda^2 - 5\lambda + 6) = 0 \quad \lambda_1 = 3 \quad \lambda_2 = 2$$

$$y_d(n) = C_1 \cdot (3)^n + C_2 \cdot (2)^n$$

$$\begin{cases} n=0 & y(0) - 5y(-1) + 6y(-2) = 0 & y(0) = -1 \\ n=1 & y(1) - 5y(0) + 6y(-1) = 0 & y(1) = -11 \end{cases}$$

$$y_d(0) = C_1 + C_2 = -1 \rightarrow -2C_1 - 2C_2 = 2$$

$$y_d(1) = 3C_1 + 2C_2 = -11 \rightarrow 3C_1 + 2C_2 = -11$$

$$C_1 = -9$$

$$C_2 = 8$$

$$y_d(n) = -9 \cdot (3)^n + 8 \cdot (2)^n$$

b) Zorlanmış Çözüm ($y_z(n)$)

$$y_z(n) = y_d(n) + y_o(n)$$

$$y_o(n) = K_1 \cdot \cos \frac{n\pi}{2} + K_2 \cdot \sin \frac{n\pi}{2}$$

$$y(n) - 5y(n-1) + 6y(n-2) = x(n)$$

$$K_1 \cdot \cos \frac{n\pi}{2} + K_2 \cdot \sin \frac{n\pi}{2} - 5 \left[K_1 \cdot \cos \left(\frac{n\pi}{2} - \frac{\pi}{2} \right) + K_2 \cdot \sin \left(\frac{n\pi}{2} - \frac{\pi}{2} \right) \right] + 6 \left[K_1 \cdot \cos \left(\frac{n\pi}{2} - \pi \right) + K_2 \cdot \sin \left(\frac{n\pi}{2} - \pi \right) \right] = \cos \frac{n\pi}{2}$$

$$K_1 \cdot \cos \frac{n\pi}{2} + K_2 \cdot \sin \frac{n\pi}{2} - 5 \left[K_1 \cdot \sin \frac{n\pi}{2} - K_2 \cdot \cos \frac{n\pi}{2} \right] + 6 \left[-K_1 \cdot \cos \frac{n\pi}{2} - K_2 \cdot \sin \frac{n\pi}{2} \right] = \cos \frac{n\pi}{2}$$

$$K_1 \cdot \cos \frac{n\pi}{2} + K_2 \cdot \sin \frac{n\pi}{2} - 5K_1 \cdot \sin \frac{n\pi}{2} + 5K_2 \cdot \cos \frac{n\pi}{2} - 6K_1 \cdot \cos \frac{n\pi}{2} - 6K_2 \cdot \sin \frac{n\pi}{2} = \cos \frac{n\pi}{2}$$

$$\cos \frac{n\pi}{2} (K_1 + 5K_2 - 6K_1) + \sin \frac{n\pi}{2} (K_2 - 5K_1 - 6K_2) = \cos \frac{n\pi}{2}$$

$$-5K_1 - 5K_2 = 0$$

$$-5K_1 + 5K_2 = 1$$

$$-10K_1 = 1$$

$$K_1 = -0.1$$

$$K_2 = 0.1$$

$$y_z(n) = C_1 \cdot (3)^n + C_2 \cdot (2)^n - (0.1) \cdot \cos \frac{n\pi}{2} + (0.1) \cdot \sin \frac{n\pi}{2}$$

$$\begin{cases} n=0 & y(0) - 5y(-1) + 6y(-2) = 1 & y(0) = 1 \\ n=1 & y(1) - 5y(0) + 6y(-1) = 0 & y(1) = 5 \end{cases}$$

$$y_z(0) = C_1 + C_2 - 0.1 = 1 \quad C_1 + C_2 = 1.1 \quad y_z(1) = 3C_1 + 2C_2 + 0.1 = 5 \quad 3C_1 + 2C_2 = 4.9$$

$$-2C_1 - 2C_2 = -2.2$$

$$3C_1 + 2C_2 = 4.9$$

$$C_1 = 2.7$$

$$C_2 = -1.6$$

$$y_T(n) = (2.7) \cdot 3^n - (1.6) \cdot 2^n - (0.1) \cos \frac{n\pi}{2} + (0.1) \sin \frac{n\pi}{2} - 9 \cdot (3)^n + 8 \cdot (2)^n$$

3. Soru $y(n) - 6y(n-1) + 9y(n-2) = x(n)$ $y(-1) = 2$ $y(-2) = 4$

$\rightarrow x(n) = \delta(n)$

a) Doğal Gözüm

$$y(n) - 6y(n-1) + 9y(n-2) = 0 \quad y(n) = \lambda^n \Rightarrow \lambda^n - 6\lambda^{n-1} + 9\lambda^{n-2} = 0 \quad \lambda^{n-2}(\lambda^2 - 6\lambda + 9) = 0$$

$$y_d(n) = C_1(3)^n + C_2n(3)^n \quad \lambda_1 = \lambda_2 = 3$$

b) Zorlanmış Gözüm

$$y_2(n) = y_d(n) + y_h(n) \quad y_2(n) = y_d(n) = C_1(3)^n + C_2n(3)^n$$

$$n=0 \quad y(0) - 6y(-1) + 9y(-2) = 1 \quad y(0) = 1$$

$$n=1 \quad y(1) - 6y(0) + 9y(-1) = 0 \quad y(1) = 6$$

$$y_2(0) = C_1 = 1$$

$$y_2(1) = 3C_1 + 2C_2 = 6$$

$$C_2 = 1$$

$$h(n) = y_2(n) = (3)^n + n(3)^n$$

5. Soru

$$y(n) - 3y(n-1) - 4y(n-2) = x(n) + 2x(n-1)$$

$$h(n) = ?$$

$$X(z) \cdot H(z) = Y(z) \quad Y(z)/X(z) = H(z)$$

$$\frac{Y(z)}{X(z)} = \frac{1 + 2z^{-1}}{1 - 3z^{-1} - 4z^{-2}} = H(z)$$

$$Y(z) - 3z^{-1}Y(z) - 4z^{-2}Y(z) = X(z) + 2z^{-1}X(z)$$

$$Y(z)[1 - 3z^{-1} - 4z^{-2}] = X(z)[1 + 2z^{-1}]$$

$$H(z) = \frac{1 + 2z^{-1}}{z^{-2}(z^2 - 3z - 4)} = \frac{z^2 + 2z}{(z-4)(z+1)}$$

$$H(z) \cdot z^{n-1} = \frac{z^{n+1} + 2z^n}{(z-4)(z+1)}$$

$$H(z) \cdot z^{n-1} (z-4) = \frac{z^{n+1} + 2z^n}{(z+1)} \Big|_{z=4} = \frac{4^{n+1} + 2 \cdot 4^n}{5}$$

$$H(z) \cdot z^{n-1} (z+1) = \frac{z^{n+1} + 2z^n}{(z-4)} \Big|_{z=-1} = \frac{(-1)^{n+1} + 2(-1)^n}{-5}$$

$$h(n) = \frac{4 \cdot 4^n + 2 \cdot 4^n}{5} + \frac{(-1) \cdot (-1)^n + 2 \cdot (-1)^n}{-5} = \frac{4^n(4+2)}{5} + \frac{(-1)^n(-1+2)}{-5} = \frac{6 \cdot 4^n}{5} - \frac{(-1)^n}{5}$$

$$h(n) = \frac{1}{5} (6 \cdot 4^n - (-1)^n) \cdot u(n)$$

$$\lambda(n) = L_n^A \cdot u(n)$$

$$y(-1) = 0 \quad y(-2) = 0$$

$$g(x) = x^2$$

$$\lambda^2 - 3\lambda + 1 = 4\lambda^2 - 1 = 0$$

$$\lambda_1 = 4 \quad \lambda_2 = -1$$

$$g_d(n) = C_1 \cdot 4^n + C_2 \cdot (-1)^n$$

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$$y(0) - 2y(-1) + 4y(-2) = 0$$

$$y(t) = 0$$

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$$y(1) = 3y(0) = 4y(-1) = 0$$

$$y(1) = 0$$

$$C_1 = 0 \quad C_2 = 0$$

$$y_d(n) = 0$$

$$x(n) = 4^n \cdot u(n)$$

$$30(n) = K_1 \alpha_1^2(n)$$

$$\longleftrightarrow y_0(n) = K \cdot B^n \cdot u(n)$$

$$22 \quad K \cdot n \cdot 4^n - \frac{2}{4} \cdot K \cdot (n-1) \cdot 4^n - \frac{1}{4} \cdot K \cdot (n-2) \cdot 4^n = 4^n + \frac{2}{4} \cdot 4^n$$

$$\frac{4K \cdot n \cdot 4^n - 3K \cdot (n-1) \cdot 4^n - K \cdot (n-2) \cdot 4^n}{4} = \frac{4 \cdot 4^n + 2 \cdot 4^n}{4}$$

$$4^2(4K_n - 3K_n + 3K - K_n + 2K) = 4^2(4+2)$$

$$5K = 6 \quad K = 6/5$$

$$y_0(n) = \frac{6}{5} \cdot n \cdot 4^n \cdot u(n)$$

$$y_E(n) = C_1 \cdot (4)^n + C_2 \cdot (-1)^n + \frac{6}{5} \cdot n \cdot 4^n \cdot u(n)$$

$$n=0$$

$$y(0) - \frac{3y(-1)}{0} - \frac{4y(-2)}{0} = 1$$

$$y(0) = 1$$

$n=1$ 1912

$$y(1) - \underbrace{3y(0)}_1 - \underbrace{4y(-1)}_9 = 4 + 2$$

$$y(1) = 2$$

$$y_2(0) = C_1 + C_2 = 1$$

$$y_2(1) = 4C_1 - C_2 + \frac{24}{5} = 9$$

$$4c_1 - c_2 = \frac{21}{5}$$

$$+ \quad C_1 + C_2 = 1$$

$$SC_1 = \frac{26}{5}$$

$$C_1 = \frac{26}{25}$$

$$C_2 = -\frac{1}{2}$$

$$y_T(n) = y_2(n) + \underbrace{y_d(n)}_0 = \frac{26}{25} (4)^n + \frac{1}{25} (-1)^n + \frac{6}{5} \cdot n \cdot 4^n \cdot u(n)$$

6. Sign $x(k) + \frac{1}{3}x(k-1) = u(k) + u(k-1) \quad k \geq 0 \quad u(k) = \left(\frac{1}{3}\right)^k$

a) $X(z) = \sum_{k=-\infty}^{\infty} x(k) \cdot z^{-n}$

$X(z) + \frac{1}{3}z^{-1}X(z) = U(z) + z^{-1}U(z)$

$u(k) = \left(\frac{1}{3}\right)^k$

$U(z) = \sum_{k=-\infty}^{\infty} \left(\frac{1}{3}\right)^k \cdot z^{-n} = \sum_{k=0}^{\infty} \left(\frac{1}{3}\right)^k \cdot z^{-n} = \sum_{k=0}^{\infty} \left(\frac{1}{3} \cdot z^{-1}\right)^k = \frac{1}{1 - \frac{1}{3}z^{-1}}$

$X(z) \left(1 + \frac{1}{3}z^{-1}\right) = U(z) (1 + z^{-1})$

$X(z) \left(1 + \frac{1}{3}z^{-1}\right) = \frac{1}{1 - \frac{1}{3}z^{-1}} \cdot (1 + z^{-1})$

$X(z) = \frac{1 + z^{-1}}{\left(1 - \frac{1}{3}z^{-1}\right) \left(1 + \frac{1}{3}z^{-1}\right)}$

b) $X(z) + \frac{1}{3} \left(\sum_{k=0}^{\infty} x(k-1) \cdot z^{-n} \right) = U(z) + \sum_{k=0}^{\infty} u(k-1) \cdot z^{-n}$

$\rightarrow x(-1) + x(-1) \cdot z^{-1}$

$x(-1) + z^{-1}x(z)$

$3 + z^{-1}X(z)$

$\rightarrow u(-1) + u(0) \cdot z^{-1}$

$u(-1) + z^{-1}u(z)$

$0 + z^{-1}U(z)$

$\sum_{k=0}^{\infty} u(k-1) \cdot z^{-n} = u(-1) + u(0) \cdot z^{-1}$
 $= u(-1) + z^{-1}U(z)$

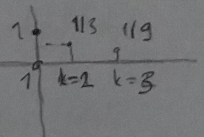
$X(z) + 1 + \frac{1}{3}z^{-1}X(z) = U(z) + z^{-1}U(z)$

$X(z) \left(1 + \frac{1}{3}z^{-1}\right) = U(z) (1 + z^{-1}) - 1$

$X(z) = \frac{(1 + z^{-1})}{\left(1 + \frac{1}{3}z^{-1}\right)} \cdot U(z) - \frac{1}{\left(1 + \frac{1}{3}z^{-1}\right)}$

c) $x(k+1) + \frac{1}{3}x(k) = u(k+1) + u(k) \quad k+1 \geq 0$
 $k \geq -1$

$u(k) = \left(\frac{1}{3}\right)^k$



$zX(z) + \frac{1}{3}X(z) = zU(z) + U(z)$

$X(z) \left(z + \frac{1}{3}\right) = U(z) (z + 1) - z$

$U(z) = \sum_{k=-1}^{\infty} \left(\frac{1}{3}\right)^{k+1} \cdot z^{-k} = \sum_{k=0}^{\infty} \left(\frac{1}{3}\right) \cdot \left(\frac{1}{3}\right)^k \cdot z^{-k} + z = \frac{1}{3} \sum_{k=0}^{\infty} \left(\frac{1}{3} \cdot z^{-1}\right)^k + z = \frac{1}{3} \left[\frac{1}{1 - \frac{1}{3}z^{-1}} \right] + z$

$(1 - z)$

$\frac{(z+1)}{1 - \frac{1}{3}z^{-1}} - z = \frac{(z+1) + (-z) \left(\frac{1}{3}\right)}{1 - \frac{1}{3}z^{-1}}$

$\left(z + \frac{1/3}{1 - \frac{1}{3}z^{-1}}\right) (z+1)$

1. Soru Fourier Dönüşümlerini Yapınız

$$x(\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} dt \rightarrow x(t) \text{ sarkelli ise}$$

a) $x(t) = e^{at} \cdot u(-t)$

$$x(\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} dt = \int_{-\infty}^0 e^{at} \cdot e^{-j\omega t} dt = \int_{-\infty}^0 e^{t(a-j\omega)} dt = \left. \frac{e^{t(a-j\omega)}}{a-j\omega} \right|_{-\infty}^0 = \frac{1}{a-j\omega} - 0 = \frac{1}{a-j\omega}$$

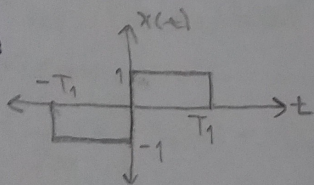
b) $x(t) = -u(t+1) + 2u(t) - u(t-1)$

$$\begin{aligned} x(\omega) &= \int_{-\infty}^{\infty} [-u(t+1) + 2u(t) - u(t-1)] \cdot e^{-j\omega t} dt = -\int_{-1}^{\infty} e^{-j\omega t} dt + 2 \cdot \int_0^{\infty} e^{-j\omega t} dt - \int_1^{\infty} e^{-j\omega t} dt \\ &= -\left. \frac{e^{-j\omega t}}{j\omega} \right|_{-1}^{\infty} - 2 \cdot \left. \frac{e^{-j\omega t}}{j\omega} \right|_0^{\infty} + \left. \frac{e^{-j\omega t}}{j\omega} \right|_1^{\infty} \\ &= -\frac{e^{j\omega}}{j\omega} + 2 \cdot \frac{1}{j\omega} - \frac{e^{-j\omega}}{j\omega} = \frac{2}{j\omega} - \frac{2}{j\omega} \left(\frac{e^{j\omega} + e^{-j\omega}}{2} \right) = \frac{2}{j\omega} (1 - \cos \omega) \end{aligned}$$

c) $x(t) = -2u(t+1) - 2u(t-1)$

$$\begin{aligned} x(\omega) &= \int_{-\infty}^{\infty} (-2u(t+1)) \cdot e^{-j\omega t} dt + \int_{-\infty}^{\infty} (-2u(t-1)) \cdot e^{-j\omega t} dt = -2 \int_{-1}^{\infty} e^{-j\omega t} dt - 2 \cdot \int_1^{\infty} e^{-j\omega t} dt \\ &= +2 \cdot \left. \frac{e^{-j\omega t}}{j\omega} \right|_{-1}^{\infty} + 2 \cdot \left. \frac{e^{-j\omega t}}{j\omega} \right|_1^{\infty} = -2 \cdot \frac{e^{j\omega}}{j\omega} - 2 \cdot \frac{e^{-j\omega}}{j\omega} = -\frac{4}{j\omega} \left(\frac{e^{j\omega} + e^{-j\omega}}{2} \right) = -\frac{4}{j\omega} \cos \omega \end{aligned}$$

2. Soru



Sarkelli zaman isaretin fourier dönüşümü?

$$x(\omega) = \int_{-T_1}^{T_1} x(t) \cdot e^{-j\omega t} dt = -\int_{-T_1}^0 e^{-j\omega t} dt + \int_0^{T_1} e^{-j\omega t} dt$$

$$\begin{aligned} x(\omega) &= +\left. \frac{e^{-j\omega t}}{j\omega} \right|_{-T_1}^0 - \left. \frac{e^{-j\omega t}}{j\omega} \right|_0^{T_1} = \frac{1}{j\omega} - \frac{e^{j\omega T_1}}{j\omega} - \frac{e^{-j\omega T_1}}{j\omega} + \frac{1}{j\omega} = \frac{2}{j\omega} - \frac{2}{j\omega} \left(\frac{e^{j\omega T_1} + e^{-j\omega T_1}}{2} \right) \\ &= \frac{2}{j\omega} (1 - \cos(\omega T_1)) \end{aligned}$$

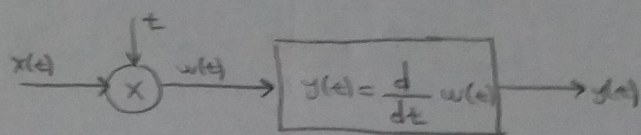
3. Soru

$$x(t) = \begin{cases} 0 & |t| > T_1 \\ \cos \pi t & |t| \leq T_1 \end{cases}$$

$x(t)$ isaretinin fourier dönüşümü?

$$\begin{aligned} x(\omega) &= \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} dt = \int_{-T_1}^{T_1} \cos \pi t \cdot e^{-j\omega t} dt = \int_{-T_1}^{T_1} \frac{e^{j\pi t} + e^{-j\pi t}}{2} \cdot e^{-j\omega t} dt = \frac{1}{2} \int_{-T_1}^{T_1} e^{t(j\pi - j\omega)} dt + \frac{1}{2} \int_{-T_1}^{T_1} e^{t(-j\pi - j\omega)} dt \\ &= \frac{1}{2} \cdot \left. \frac{e^{t(j\pi - j\omega)}}{j\pi - j\omega} \right|_{-T_1}^{T_1} - \frac{1}{2} \cdot \left. \frac{e^{t(-j\pi - j\omega)}}{j\pi + j\omega} \right|_{-T_1}^{T_1} = \frac{1}{2} \left(\frac{e^{T_1(j\pi - j\omega)}}{j\pi - j\omega} - \frac{e^{-T_1(j\pi - j\omega)}}{j\pi - j\omega} \right) - \frac{1}{2} \left(\frac{e^{T_1(-j\pi - j\omega)}}{j\pi + j\omega} - \frac{e^{-T_1(-j\pi - j\omega)}}{j\pi + j\omega} \right) \\ &= \frac{1}{(\pi - \omega)} \sin(\pi - \omega) \cdot T_1 - \frac{1}{(\pi + \omega)} \sin(-\pi - \omega) \cdot T_1 = \frac{1}{2} \left(\frac{e^{T_1(j\pi - j\omega)}}{j\pi - j\omega} - \frac{e^{-T_1(j\pi - j\omega)}}{j\pi - j\omega} \right) - \frac{1}{2} \left(\frac{e^{T_1(-j\pi - j\omega)}}{j\pi + j\omega} - \frac{e^{-T_1(-j\pi - j\omega)}}{j\pi + j\omega} \right) \end{aligned}$$

4. Soru



Sürekli zaman $x(t)$ 'nin fourier dönüşümünün $X(\omega)$ olduğu bilindiği üzere aşağıdaki sistemin elde edilen $y(t)$ işaretinin fourier dönüşümü $Y(\omega)$ nedir?

$$w(t) = t \cdot x(t)$$

$$W(\omega) = F(t \cdot x(t)) = j \cdot \frac{dX(\omega)}{d\omega}$$

$$y(t) = \frac{d}{dt} w(t)$$

$$Y(\omega) = j \cdot \omega \cdot W(\omega)$$

$$= j \cdot \omega \cdot j \cdot \frac{dX(\omega)}{d\omega} = -\omega \cdot \frac{dX(\omega)}{d\omega}$$

$$F(t \cdot x(t)) = j \cdot \frac{dX(\omega)}{d\omega} \quad f\left[\frac{dx(t)}{dt}\right] = j \cdot \omega X(\omega)$$

5. Soru

Frekans spektrumu $2\pi\delta(\omega - \omega_0)$ şeklinde verilen sürekli zaman işaretini ters fourier dönüşümü ile bulunuz.

$$X(\omega) = 2\pi\delta(\omega - \omega_0)$$

$$x(t) = ?$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} 2\pi\delta(\omega - \omega_0) \cdot e^{j\omega t} \cdot d\omega = e^{j\omega_0 t}$$

6. Soru

Frekans spektrumu $X(\omega) = \pi[\delta(\omega - 6\pi) + \delta(\omega - 4\pi) + \delta(\omega + 4\pi) + \delta(\omega + 6\pi)]$ şeklinde verilen periyodik işaretin

a) Tutarlı frekansı

b) Fourier seri katsayısını

c) $x(t)$ 'yi bulunuz?

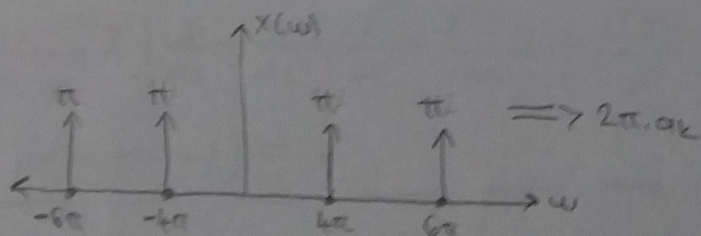
$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \pi\delta(\omega - 6\pi) \cdot e^{j\omega t} \cdot d\omega + \frac{1}{2\pi} \int_{-\infty}^{\infty} \pi\delta(\omega - 4\pi) \cdot e^{j\omega t} \cdot d\omega + \frac{1}{2\pi} \int_{-\infty}^{\infty} \pi\delta(\omega + 4\pi) \cdot e^{j\omega t} \cdot d\omega + \frac{1}{2\pi} \int_{-\infty}^{\infty} \pi\delta(\omega + 6\pi) \cdot e^{j\omega t} \cdot d\omega$$

$$= \frac{1}{2} \cdot e^{j6\pi t} + \frac{1}{2} \cdot e^{j4\pi t} + \frac{1}{2} \cdot e^{-j4\pi t} + \frac{1}{2} \cdot e^{-j6\pi t}$$

$$EBOB(4\pi, 6\pi) = 2\pi$$

$$\omega_0 = 2\pi$$

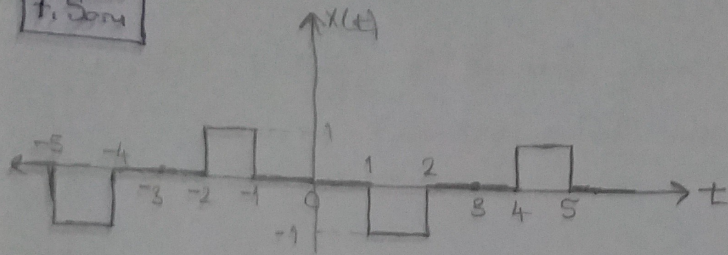
$$= \cos(6\pi t) + \cos(4\pi t)$$



$$a_3 = a_{-3} = \frac{1}{2}$$

$$a_2 = a_{-2} = \frac{1}{2}$$

7. Soru



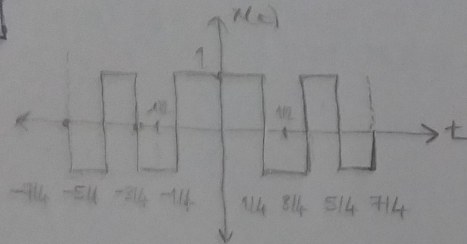
Verilen $x(t)$ periyodik işaretin Fourier serisini (katsayısını) bulunuz?

$$a_k = \frac{1}{P} \int_{-P/2}^{P/2} x(t) \cdot e^{-jk\omega_0 t} dt$$

$$P=6 \quad \omega_0 = \frac{2\pi}{P} = \frac{2\pi}{6} = \frac{\pi}{3}$$

$$\begin{aligned} a_k &= \frac{1}{6} \int_{-3}^3 x(t) \cdot e^{-jk\omega_0 t} dt = \frac{1}{6} \left[\int_{-2}^{-1} e^{-jk\omega_0 t} dt - \int_{1}^2 e^{-jk\omega_0 t} dt \right] = \frac{1}{6} \left[-\frac{e^{-jk\omega_0 t}}{jk\omega_0} \Big|_{-2}^{-1} + \frac{e^{-jk\omega_0 t}}{jk\omega_0} \Big|_1^2 \right] \\ &= \frac{1}{6} \left[-\frac{e^{-jk\omega_0 1}}{jk\omega_0} + \frac{e^{-jk\omega_0 2}}{jk\omega_0} + \frac{e^{-jk\omega_0 2}}{jk\omega_0} - \frac{e^{-jk\omega_0 1}}{jk\omega_0} \right] = \frac{2}{6jk\omega_0} \left[-\left(\frac{e^{-jk\omega_0 1} + e^{-jk\omega_0 2}}{2} \right) + \left(\frac{e^{-jk\omega_0 2} + e^{-jk\omega_0 1}}{2} \right) \right] \\ &= \frac{1}{3jk\omega_0} \left(\cos(2k\omega_0) - \cos(k\omega_0) \right) = \frac{1}{jk\pi} \left(\cos\left(\frac{2k\pi}{3}\right) - \cos\left(\frac{k\pi}{3}\right) \right) \end{aligned}$$

8. Soru



Verilen $x(t)$ periyodik işaretin Fourier serisini (katsayısını) bulunuz.

$$P = \frac{1}{2} - \frac{0}{2} = \frac{1}{2}$$

$$\omega_0 = 2\pi$$

$$a_k = \frac{1}{P} \int_{-P/2}^{P/2} x(t) \cdot e^{-jk\omega_0 t} dt$$

$$a_k = -\int_{-1/2}^{-1/4} e^{-jk\omega_0 t} dt + \int_{1/4}^{1/2} e^{-jk\omega_0 t} dt = \int_{1/4}^{1/2} e^{-jk\omega_0 t} dt = \frac{e^{-jk\omega_0 t}}{-jk\omega_0} \Big|_{1/4}^{1/2} = \frac{e^{-jk\omega_0 1/2}}{-jk\omega_0} - \frac{e^{-jk\omega_0 1/4}}{-jk\omega_0} = \frac{e^{-jk\omega_0 1/2} - e^{-jk\omega_0 1/4}}{-jk\omega_0}$$

$$a_k = \frac{2}{k\omega_0} \left[2 \sin \frac{k\omega_0}{4} - \sin \frac{k\omega_0}{2} \right] = \frac{1}{k\pi} \left[2 \sin \frac{k\pi}{2} - \sin k\pi \right]$$

9. Soru

11. Soru

$f_0 = 1 \text{ kHz}$ olarak üzere $x(t) = \sin \omega_0 t$ analog işaretinin $T_c = \frac{1}{2100}$ sn aralıklarla örneklendiği ile elde edilen $x(nT)$ işaretinin frekans spektrumu?

$$\omega_0 = 2\pi f_0$$

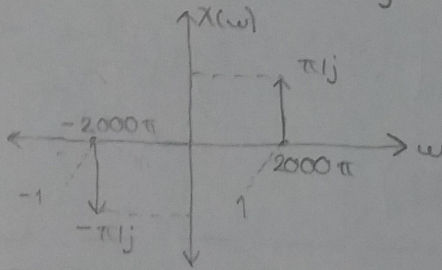
$$X_s(\omega) = \frac{1}{2\pi} [X_a(\omega) * S(\omega)]$$

Bu analog işaretin belirli aralıklarla örneklenecek için kullanılacak formül

$$S(\omega) = \sum_{k=-\infty}^{\infty} \omega_c \cdot \delta(\omega - k \cdot \omega_c) \quad \omega_c = \omega_c \quad T_c \text{ aralıklarla örnekleme}$$

$$X_a(t) = \sin \omega_0 t = \frac{e^{j\omega_0 t} - e^{-j\omega_0 t}}{2j}$$

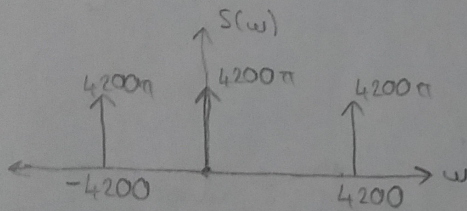
$$\omega_0 = 2\pi \cdot 1000 = 2000\pi$$



$$S(\omega) = \sum_{k=-1}^1 \omega_c \cdot \delta(\omega - k \cdot \omega_c)$$

$$\omega_c = \frac{2\pi}{T_c} = 2\pi \cdot 2100 = 4200\pi$$

$$= \sum_{k=-1}^1 4200\pi \cdot \delta(\omega - 4200k\pi)$$



$$X_s(\omega) = \frac{1}{2\pi} \cdot [X_a(\omega) * S(\omega)]$$

$$X_s(\omega) = \frac{1}{2\pi} \left[\begin{array}{c} \text{Plot of } X_a(\omega) \text{ (impulses at } \pm 2000\pi \text{)} \\ * \\ \text{Plot of } S(\omega) \text{ (impulses at } -4200, 0, 4200 \text{)} \end{array} \right]$$

